

Matrix Multiplication and Vectors

ECON 441: Introduction to Mathematical Economics
Lecture 4

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Matrix Multiplication

A whole new animal...

Only possible to multiply two matrices, $A_{m \times n}$ and $B_{p \times q}$ to get AB if $n = p$ i.e.

number of columns in A = number of rows in B

Example. $A = \begin{bmatrix} 2 & 3 & 1 \\ 4 & -6 & -2 \end{bmatrix}_{2 \times 3}$ $B = \begin{bmatrix} 1 & 8 \\ -2 & 3 \end{bmatrix}_{2 \times 2}$

Cannot do AB , but can do BA

Matrix Multiplication

Another example.

$$A = \begin{bmatrix} 2 & 3 & 1 \\ 4 & -6 & -2 \end{bmatrix}_{2 \times 3} \quad B = \begin{bmatrix} 1 \\ -2 \\ 4 \end{bmatrix}_{3 \times 1}$$

Can we multiply A and B to find $C = AB$?

Yes, since A has 3 columns, which is equal to the number of rows in B .

Also, the dimension of C will be 2×1 .

Matrix Multiplication

So how to actually multiply these matrices?

$$C = AB$$

$$c_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + \dots + a_{in}b_{nj} = \sum_{k=1}^n a_{ik}b_{kj}$$

The **element** c_{ij} is obtained by multiplying term-by-term the entries of the **i th row of A** and **j th column of B** .

Matrix Multiplication: Examples

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix}_{2 \times 3} \quad B = \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \\ b_{31} & b_{32} \end{bmatrix}_{3 \times 2}$$

Here,

$$C = AB = \begin{bmatrix} a_{11}b_{11} + a_{12}b_{21} + a_{13}b_{31} & a_{11}b_{12} + a_{12}b_{22} + a_{13}b_{32} \\ a_{21}b_{11} + a_{22}b_{21} + a_{23}b_{31} & a_{21}b_{12} + a_{22}b_{22} + a_{23}b_{32} \end{bmatrix}_{2 \times 2}$$

Matrix Multiplication: Examples

$$A = \begin{bmatrix} 2 & 3 & 1 \\ 4 & -6 & -2 \end{bmatrix}_{2 \times 3} \quad B = \begin{bmatrix} 1 \\ -2 \\ 4 \end{bmatrix}_{3 \times 1}$$

$$C = AB = \begin{bmatrix} 2 \times 1 + 3 \times -2 + 1 \times 4 \\ 4 \times 1 + -6 \times -2 + -2 \times 4 \end{bmatrix}_{2 \times 1} = \begin{bmatrix} 0 \\ 8 \end{bmatrix}_{2 \times 1}$$

Matrix Multiplication: Examples

$$A = \begin{bmatrix} 1 & 3 \\ 2 & 8 \\ 4 & 0 \end{bmatrix} \quad B = \begin{bmatrix} 5 \\ 9 \end{bmatrix}$$

A Simple Economic Model

$$A = \begin{bmatrix} 1 & 2 \\ 1 & -3 \end{bmatrix} \quad x = \begin{bmatrix} q \\ p \end{bmatrix} \quad b = \begin{bmatrix} 100 \\ 20 \end{bmatrix}$$

What is Ax ?

$$Ax = \begin{bmatrix} q + 2p \\ q - 3p \end{bmatrix}$$

Setting $Ax = b$ gives us back our demand and supply equations.

Vectors

Matrices with only one column: column vectors

$$x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

Matrices with only one row: row vectors

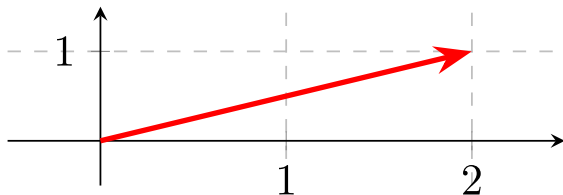
$$x' = \begin{bmatrix} x_1 & x_2 & \cdots & x_n \end{bmatrix}$$

Geometric Representation

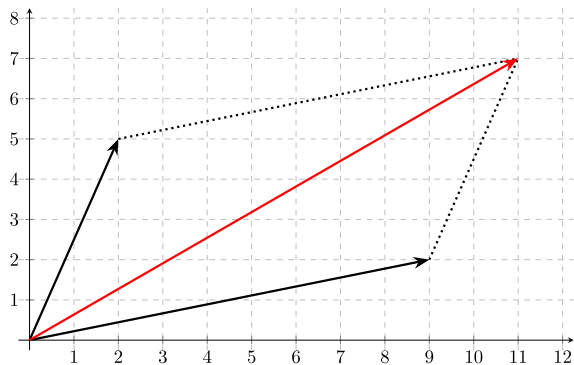
Vectors can be graphically represented by arrows

The direction of the arrow indicates the vector's **direction** and the length of the arrow corresponds to its **magnitude**.

Here is a plot for vector $v_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$



Addition of Vectors



Inner Product

Inner product of two vectors each with n elements:

$$u \cdot v = u_1 v_1 + u_2 v_2 + \dots + u_n v_n = \sum_{i=1}^n u_i v_i$$

Example.

$$u = \begin{bmatrix} 1 \\ 5 \\ 2 \end{bmatrix} \quad v = \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix}$$

Linear Dependence

A set of vectors is said to be **linearly dependent** if and only if any one of them can be expressed as a linear combination of the remaining vectors.

Example.

$$v_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \quad v_2 = \begin{bmatrix} 2 \\ 4 \end{bmatrix}$$

Linear Dependence

A set of vectors is said to be **linearly dependent** if and only if any one of them can be expressed as a linear combination of the remaining vectors.

Example.

$$v_1 = \begin{bmatrix} 3 \\ 2 \end{bmatrix} \quad v_2 = \begin{bmatrix} 1 \\ 3 \end{bmatrix} \quad v_3 = \begin{bmatrix} 1 \\ -4 \end{bmatrix}$$

Linear Dependence

A set of m -vectors $v_1, v_2, \dots, v_n$ is linearly dependent if and only if there exists a set of scalar $k_1, k_2, \dots, k_n$ (not all zero) such that:

$$\sum_{i=1}^n k_i v_i = 0 \quad (m \times 1)$$